

Project Design Phase
Proposed Solution Template

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Students preparing for exams have large amounts of study material scattered across PDFs, notes, textbooks and articles, and limited time. Summarising content, making flashcards and quizzes, and planning revision by hand is slow, repetitive and inconsistent, and there is no single secure place to store and revisit these resources.
2.	Idea / Solution description	AI StudyBuddy is a secure RESTful backend (Node.js, Express.js, MongoDB with Mongoose) that connects to the Google Gemini API. After JWT login, a student uploads or pastes study material and the system generates a concise summary, flashcards, a multiple-choice quiz and a personalised study plan based on the exam date. Everything is stored per user for later revision, and administrators manage users and monitor AI usage through role-based access control.
3.	Novelty / Uniqueness	Content is generated dynamically by Gemini from the student's own material instead of fixed templates. Summaries, flashcards, quizzes and study plans come from one platform, so students do not switch between apps. A modular AI service layer keeps the AI model replaceable, and results are saved and linked to each material. Security is built in through JWT, bcrypt and RBAC.
4.	Social Impact / Customer Satisfaction	Cuts manual preparation time so students can focus on understanding concepts. Gives students who cannot afford private tutoring access to personalised revision support. Active-recall tools (flashcards, quizzes) and a clear plan reduce exam stress and improve confidence. Consistent, organised resources improve satisfaction and learning outcomes.

5.	Business Model (Revenue Model)	Freemium: a free tier with a daily limit on AI generations, and a premium subscription for higher limits, larger uploads and advanced study plans. Institutional licences for colleges and coaching centres, including the admin dashboard and usage reports.
6.	Scalability of the Solution	The stateless JWT-based REST API can scale horizontally behind a load balancer. MongoDB supports replica sets and sharding as data grows. AI calls can be rate-limited, cached or queued to control cost and latency. The modular MVC design lets new AI features (for example mind maps or voice notes) be added with minimal changes.

References:

Node.js Documentation: <https://nodejs.org/en/docs>

Mongoose ODM Documentation: <https://mongoosejs.com/docs/>

JSON Web Token (JWT) Introduction: <https://jwt.io/introduction>

Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>

React Documentation: <https://react.dev/>

Postman Learning Center: <https://learning.postman.com/docs/>